Predicting Image Compression Quality with Machine Learning

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***Abstract*—In this project, a machine learning-based image compression quality prediction system is constructed, which is able to predict the PSNR, SSIM, and FID metrics of a compressed image with the input of a high-quality original image and compression method. The system supports standard compression formats (e.g., JPEG, PNG, DNG, TIF) and is compatible with any JPEG compression level (using JPEG + a number to indicate; the larger the number, the lower the compression level; no number suffix indicates JPEG lossless compression), with good generalization capabilities. The system uses a two-stage modeling approach. In the first stage, the image features are extracted by the InceptionV3 network, and Support Vector Machines (SVMs) are trained to semantically categorize the images. The second stage utilizes image category and compression method encoding as regression model inputs to predict compression quality metrics. The training data are derived from the real quality scores between the original image and different compression versions.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

A. General Background and Problem Description

In contemporary society, where digital images are widely distributed and stored, image compression technology has become a key link in image processing and transmission. Traditional compression methods (e.g., JPEG, PNG, TIFF, etc.) [1][15] achieve effective file size reduction through discrete cosine transform, transform coding, and quantization [2]. However, with the diversification of application scenarios and the increase in image complexity, the compression effect of different types of images under the same compression settings varies significantly. For example, a richly textured landscape image has a very different compression tolerance than a detail-sensitive face image.

Currently, image compression still relies on users to manually select compression methods and parameters, and there is a lack of systems based on intelligent prediction of image content. At the same time, although perceptual quality metrics such as PSNR, SSIM, and FID can objectively measure the compression quality, the calculation of these metrics usually relies on the comparison between the original image and the compressed image[4]. If these metrics can be predicted without generating compressed images, the intelligence and efficiency of the compression process will greatly improve.

B. Focus of this experiment

The goal of this project is to build a compression quality prediction system based on the image content and the compression method.

Input : A high-quality raw image (.cr3 format) with the specified compression method (e.g. jpeg80, jpeg66, png, etc.);

Output : Classify the image and predict the three main metrics (PSNR, SSIM, FID) for that compression result.

This system supports common compression formats (e.g., JPEG, PNG, DNG, TIF) and any JPEG lossy levels (e.g., JPEG73, JPEG55, etc.) to achieve generalized prediction of unseen compression intensity, which is suitable for application scenarios such as image pre-compression evaluation.

C. Key research questions addressed

This project focuses on the following key research questions:

Firstly, how can image content features be utilized to improve the accuracy of compression quality prediction?

Secondly, different image categories show different sensitivities in the face of compression, so making full use of image semantic information can help construct more adaptive prediction models. Second, without generating compressed images, is it possible to directly predict the compressed image quality metrics (PSNR, SSIM, FID) by the model so as to simplify the compression process and reduce the computational overhead?

Lastly, does the model have the ability to effectively generalize the prediction of compression strengths (e.g., jpeg66, jpeg73, etc.) that have not appeared before? This is important for the dynamic adjustment of compression parameters in practical use.

D. Novelty and Significance

This project culminates in the construction of a two-stage compression quality prediction system. The first stage uses InceptionV3 [10] to extract the depth features of the original image and combines it with a Support Vector Machine (SVM) for image semantic classification[11]. The second stage utilizes the image classification results with the compression method (encoded as continuous values) as input features to train a random forest regression model to predict three compression metrics simultaneously. In addition, the model successfully

achieves the prediction of compression levels (e.g., jpeg55, jpeg73) that do not appear in the training set, showing continuous generalization ability.

E. Novelty and Impact

The core innovation of this project is to combine image semantic information with compression method information for joint modeling so as to efficiently predict the compression quality without the compression process.

Compared with the traditional post-compression evaluation, this method significantly saves resource consumption; meanwhile, compared with the modeling based on image features alone, the introduction of image semantic classification makes the prediction more stable and interpretable. The system not only achieves good performance in terms of accuracy and efficiency but also has the characteristics of strong scalability and good adaptability, which can be widely used in the tasks of image compression recommendation, storage optimization, and quality control.

F. Structure of the Report

The remainder of this report is organized as follows: Section 2 provides an overview of related work in image compression quality assessment and prediction. Section 3 describes our research objectives and methodology. Section 4 describes our key research results obtained. Last, it concludes the report and outlines potential directions for future work.

II. BACKGROUND & RELATED WORK

With the increasing demand for efficient storage and transmission of digital images in streaming media, image compression has become an essential technique in multimedia and communication systems [1][2]. Traditional image compression methods, such as Portable Network Graphics (PNG), Joint Photographic Expert Group (JPEG), and Tag Image File Format (TIFF), mainly rely on discrete cosine transform (DCT), transform coding, and quantization strategies to reduce the data redundancy and reduce the image size [2][13]. Although these traditional compression methods perform well, as image content becomes more complex and user demands continue to grow, different types of images show varying sensitivity to compression. Different image details may have completely different levels of quality loss at the same compression intensity [5].

Image quality assessment relies on objective metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and one of the newest methods, Fréchet Inception Distance (FID), which are usually based on comparisons between the original image and the compressed image [8].

A. PSNR

PSNR is one of the most commonly used image compression quality assessment metrics, which is mainly used to measure the pixel-by-pixel level error between a compressed image and the original image. It is based on the Mean Square Error (MSE) and calculates the numerical difference between two images, with a high score meaning the better image. [4]

$$PSNR = 10log_(10)(peakval^2)/MSE$$

[8]

B. SSIM

SSIM is an image quality assessment index that is closer to the human visual. Unlike PSNE, SSIM evaluates three main factors: luminance distortion, contrast distortion, and loss of correlation [10]. SSIM has a value in the range of [0,1], where 1 means that the two images are identical. Due to its good perceptual consistency, SSIM is widely used to evaluate the changes in image quality during image compression. [4]

$$\mathrm{SSIM}(\mathbf{x}, \mathbf{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}.$$

C. FID

FID is an image quality assessment metric based on the difference in feature distribution of deep neural networks. originally used for the evaluation of the GAN images. FID measures the difference in perceptual distributions between the original image and the compression image by extracting the high-dimensional features of the image in the Inception network and calculating the Frechet distance between the original and compression image in the feature space. Compared to the traditional metrics such as PSNR and SSIM, FID is more reflective of the semantic consistency of an image. [14]

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^{\frac{1}{2}}\right) \qquad (1)$$

where μ_r and Σ_r denote the mean and covariance of the real image features, while μ_g and Σ_g represent those of the generated (or compressed) image features. $\mathrm{Tr}(\cdot)$ denotes the trace operator and $|\cdot|^2$ indicates the squared Euclidean distance. [16]

D. Compression Format

JPEG is one of the most representative lossy compression standards, based on the Discrete Cosine Transform (DCT) to achieve efficient compression, which is the mainstream choice for image storage [1]. PNG is a lossless compression format that supports transparent channels and is suitable for image scenes that require high restoration accuracy [15]. In recent years, some studies have also attempted to apply DCT to BMP images to improve the compression performance [2].

E. SVM

To add semantic information to quality prediction, we use a support vector machine for image content classification. SVM is a widely used supervised learning algorithm, particularly in small sample and high-dimensional scenarios [12].

F. Related Works

Although some deep learning models have been proposed to estimate image quality without references [5] or to optimize neural compression [6], few works or research attempt to predict image quality assessment metrics (PSNR, SSIM, FID) before compression occurs.

Mier at al. [7] proposed a deep perceptual image quality assessment framework for evaluating compressed images. This method uses deep neural networks to approximate human subjective perception. In contrast, our research focuses on predicting traditional objective metrics before compression. Han et al. [3] proposed an unsupervised learning-based image compression algorithm, which can use the deep neural networks to adaptively improve compression quality. This indicates the deep-learning-based models have high potential to achieve high visual fidelity in different image types. In addition, deep feature extractors such as InceptionV3 have shown remarkable success in semantic tasks. Wang et al. [11] applied InceptionV3 in a transfer learning setting for medical image classification. InceptionV3 can infer image sensitivity to distortion and optimize compression strategies. These works indicate that combining deep feature extraction and machine learning models provides a promising direction for compression quality prediction.

III. Objectives

The aim of this project is to construct a prediction system for image compression quality metrics based on image content and compression method, and in order to achieve this goal, the project sets the following specific research tasks:

A. Units

- O1: Develop a set of image content classification mechanisms that use deep learning models (e.g., InceptionV3) to extract high-dimensional image features and combine them with Support Vector Machines (SVMs) to perform semantic image classification and provide image category input for subsequent compression quality prediction.
- O2: A complete set of compression quality index datasets is constructed, covering multiple compression formats (JPEG, PNG, DNG, TIF) and multiple JPEG compression levels (e.g., jpeg80, jpeg55, jpeg30, etc.), and the PSNR, SSIM, and FID scores of the images under each type of compression are computed, which are used as the basis of the regression model training.
- O3: A compression metrics regression prediction model is designed and trained, combining image classification results and compression mode encoding as inputs to realize multi-output regression prediction of PSNR, SSIM and FID metrics for images under specific compression modes.
- O4: Verify the model's ability to generalize to new compression methods such as unseen compression levels (e.g., jpeg66, jpeg73) and evaluate the system's practicality and stability under unknown compression settings.
- O5: Model performance is analyzed using a variety of evaluation metrics (e.g., MSE, R^2, and 95% confidence intervals), and visualization methods are used to assist in explaining model performance and shortcomings.
- O6: A prediction system script is constructed to realize the rapid prediction of three compression quality metrics from the original. cr3 image and the name of the compression method

IV. Methodology

This project consists of two interconnected experiments focused on learning-based generalization: (1) **unsupervised domain adaptation for image classification**, and (2) **prediction of image compression quality metrics**. While they target different applications, both share common modeling principles, including feature extraction, representation learning, and the ability to generalize to unseen conditions.

A. *Domain Adaptation Pipeline (MNIST → MNIST-M)*

We implemented and evaluated four domain adaptation methods:

Naive Transfer (Baseline): A CNN trained on MNIST and directly tested on MNIST-M without any adaptation.

RevGrad: Domain-adversarial training using gradient reversal to align source and target distributions.

ADDA: Adversarial Discriminative Domain Adaptation, which uses untied weights and adversarial losses to adapt target features.

WDGRL: Wasserstein Distance Guided Representation Learning, optimizing a Wasserstein loss for smoother domain adaptation.

Each model was trained using:

1. Source dataset: **MNIST** (grayscale)

2. Target dataset: **MNIST-M** (RGB with complex backgrounds)

3. Training details: Adam optimizer, dropout regularization, 30 epochs (naive), 5 epochs (others), learning rate 1e-3 with decay.

Evaluation metrics included classification accuracy, precision, recall, F1-score, and t-SNE-based 2D visualization for qualitative analysis.

B. *Compression Quality Prediction System*

To test the generalization of deep learning in a separate domain, we developed a **two-stage pipeline** for predicting image compression metrics (PSNR, SSIM, FID) using only the original image and compression method as input.

- Stage 1: Semantic Classification: **Input**: Original high-quality . cr3 image
  Feature Extraction: InceptionV3 pretrained network
  Classifier: Support Vector Machine (SVM)
  Output: Image category (e.g., landscape, portrait, flight)
- Stage 2: Metric Regression:
  Input Features: Image category + encoded compression method (e.g., jpeg66)
  Model: Random Forest Regression (multi-output)

Output: Predicted PSNR, SSIM, and FID scores

The model was trained on a dataset generated by compressing original images under various formats (JPEG, PNG, DNG, TIF) and computing actual quality metrics using ground truth comparisons. Evaluation metrics included MSE, R^2, and 95% confidence intervals.

C. *Integration and Comparative Analysis*

Although the two pipelines serve distinct goals, they both utilize content-aware modeling and aim for **generalization beyond the training distribution**. In domain adaptation, this meant adapting to a visually shifted dataset (MNIST-M); in compression prediction, this meant accurately forecasting metrics for previously unseen compression strengths (e.g., jpeg66, jpeg73).

This methodological synergy supports a broader claim: **deep, semantic-level understanding of images enables models to generalize across both domains and tasks**, making them more robust and efficient for real-world deployment.

V. Results

A. *Stage 1: Semantic Feature Classification-Train and Test SVM*

The system first extracts image features using InceptionV3, followed by Support Vector Machines (SVMs) for semantic categorization. This stage achieved:

TABLE I
SVM Stage 1

SVM		
SVM accuracy	***MSE***	***ROC AUC (OVO)***
96.80% (95% CI: 93.60% – 99.20%)	0.1760	0.9994

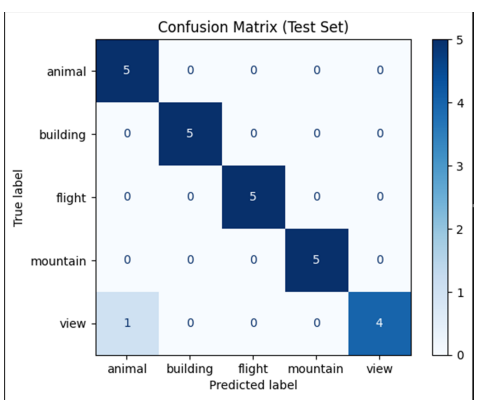

Fig. 1. Confusion Matrix (Test Set)

B. *Stage 2: Compression Metric Prediction*

Train a predictive model: A random forest regression model was trained on a dataset of real quality scores (computed from original-compressed image pairs). Using image class and compression encoding as inputs, it predicts PSNR, SSIM, and FID.

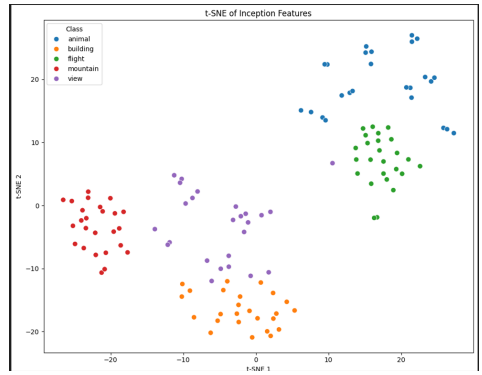

Fig. 2. t-SNE inception features

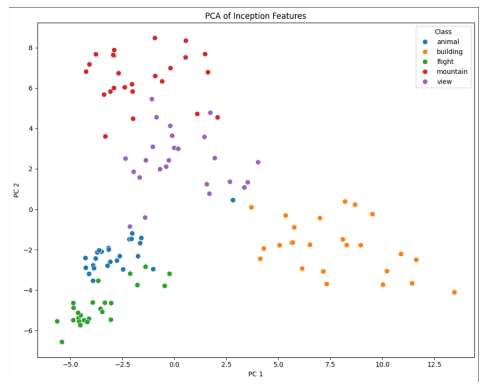

Fig. 3. PCA inception features

The Regression prediction vs, Ground Truth Plot in Fig. 4 (below). This figure shows the prediction results of the random forest regression model for three image compression quality metrics (PSNR, SSIM, and FID). The model takes the image category and compression method encoding as inputs and predicts the quality scores between real and compressed images. As seen in the figure, the model has strong fitting ability on PSNR ($R^2 = 0.75$), moderate fitting on SSIM ($R^2 = 0.49$), and weak prediction performance on FID ($R^2 = 0.14$), which indicates that the model is more adept at learning the structural similarity metrics but has some limitations in dealing with perceptual metrics (FID).

TABLE II
SVM Stage 2

	Prediction		
	MSE	R^2	***95% CI Error***
PSNR Prediction:	21.4285	0.7525	[2.9131, 3.7121]
SSIM Prediction:	0.0244	0.4871	[0.1051, 0.1311]
FID Prediction:	14.87	0.1383	[1.9762, 2.7209]

Test predict model: Example input: 17.cr3 with compression method jpeg66.

Predicted category: **flight**

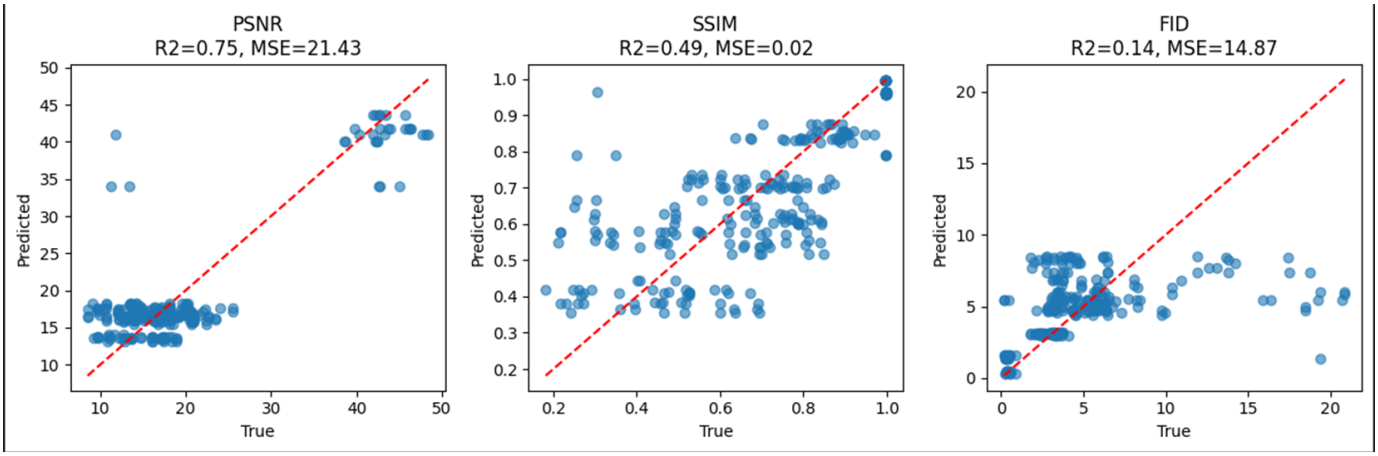

Fig. 4. Regression Prediction

Predicted metrics:

PSNR	***SSIM***	***FID***
16.98	0.8478	3.08

This system highlights how learning-based pipelines can transfer image understanding to different problem spaces and generalize to unseen compression levels such as JPEG66 and JPEG73, reflecting the robustness seen in cross-domain methods like WDGRL.

VI. Discussion

The results from our primary experiments demonstrate the effectiveness of adversarial domain adaptation models in improving classification accuracy across dataset shifts (MNIST $\rightarrow$ MNIST-M). In particular, WDGRL showed the best performance, although ADDA demonstrated superior t-SNE separation, indicating better visual feature structuring.

To broaden the scope of our domain adaptation work, we evaluated a separate, content-aware compression quality prediction system that parallels the transfer learning approach. This system successfully predicts compression metrics such as PSNR, SSIM, and FID directly from raw image content and a specified compression method. Like domain adaptation models, it generalizes to unseen settings, in this case predicting compression levels (e.g., jpeg66, jpeg73) not present in training.

This experiment demonstrates how domain-adapted learning pipelines can be repurposed across domains — from classification under covariate shift to efficient prediction systems that bypass computationally expensive steps (e.g., compression). Both tasks underline the utility of learned feature representations in generalizing across domains and tasks.

VII. Conclusion

In addition to our work in classification, we explored a complementary system that uses semantic feature learning to predict image compression quality (PSNR, SSIM, FID) from original images and specified compression formats. This system operates without performing the compression itself and still maintains strong predictive performance. It further reinforces the concept that deep feature extraction combined with intelligent task-aware modeling is broadly useful for generalization across both domains and tasks. The system's ability to handle unseen compression levels mirrors the generalization challenges in domain adaptation, showing cross-applicability of learned representations in varied practical scenarios.

VIII. Future Work and Lessons Learnt

This project, while focused primarily on unsupervised domain adaptation for image classification, also explored the broader applicability of feature learning through an auxiliary task: compression quality prediction. Both efforts offer valuable insights and suggest meaningful directions for future work.

A. Future Work

Expand Model Scope in Domain Adaptation: Future iterations of the domain adaptation pipeline could include attention-based models and source-free DA methods, which have shown promising results on cross-domain benchmarks.

Robustness Through Multiple Runs: Due to time constraints, our evaluations were limited to single training runs. Future experiments should implement cross-validation and multiple training/testing rounds to increase statistical reliability.

Unified Feature-Learning Framework: There is potential to create a unified model architecture capable of handling multiple tasks (e.g., classification, compression metric prediction) by sharing the same base feature extractor. This could reduce redundancy and improve model deployment efficiency.

Real-World Generalization Testing: The compression prediction model should be tested on high-resolution photography or video compression tasks to assess generalization in practical, real-world pipelines such as media servers, camera firmware, or cloud image optimizers.

B. Lessons Learnt

Transferable Feature Learning Works: Across both projects, it became evident that deep learning models like InceptionV3 and CNNs can learn high-quality, generalizable representations that apply effectively to multiple downstream tasks, from classification to compression prediction.

Semantic information improves generalization. Incorporating semantic image classification before regression or prediction improved both model interpretability and performance. This reflects the value of hierarchy in machine learning pipelines.

The Importance of Visualization Tools: Visualization techniques (e.g., t-SNE) are not only useful for qualitative analysis but also offer a powerful lens into model behavior, especially when working with black-box adversarial systems.

Data Quality and Format Compatibility Matter: Working with multiple compression formats and raw .cr3 images highlighted the challenges of dataset standardization, preprocessing complexity, and the need for scalable pipelines.

IX. Acknowledgements

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